

SIH Presentation vs. Actual Codebase: Comparison Report

Date: September 29, 2026
Project: MetraVerify (Legal Metrology Digital Trust Platform)

This report provides a comprehensive analysis comparing your Smart India Hackathon (SIH) PowerPoint presentation against the actual project we have built in the codebase.

1. What Matches with the Actual Project?

The core business logic and primary workflow in the PPT heavily align with the codebase we built.

- Role-Based Architecture:** The Admin Portal, Inspector (LMO), GATC, and Traders (Business) roles are all fully implemented with strict access control.
- Consumer Verification (Scan to Verify):** The public QR scanning portal (/api/public/verify/:id) is fully functional. The QR codes lead to a clean URL without leaking private data, exactly as envisioned.
- Digital Certification Engine:** The backend successfully generates and issues on-the-fly PDF certificates when an inspection passes.
- Expiration Alerts:** The expiry engine correctly categorizes instruments (Valid, Expiring Soon, Expired) and the dashboards reflect this data.
- Department Dashboards:** The frontend utilizes `Recharts` to provide live pendency views, inspection status maps, and pending task lists for Admins and LMOs.
- Digital Records & Paperless Flow:** The entire flow from application submission to certificate generation is fully digital.

2. How Different is the PPT from the Actual Project? (Discrepancies)

There is a significant gap between the **Technology Stack** and **Security Features** claimed in the PPT versus what is actually written in the code.

Feature / Tech Claimed in PPT	Actual Implementation in Codebase
NestJS & Python	Built entirely with Express.js & Node.js
PostgreSQL & PostGIS	Built using MongoDB & Mongoose (NoSQL)
React Native (Mobile App)	Built as a React 18 Web SPA (Responsive Web)
Redis, BullMQ, S3 Storage	Not implemented (Monolithic Node server architecture)
Docker & Kubernetes	Not currently containerized or orchestrated
Keycloak & DigiLocker	Custom Authentication using JWT and bcrypt
ECDsa Signed & KMS/HSM	Standard PDF generation (PDFKit) and HMAC-SHA256 tokens
Hash-chained Audit Logs	Standard MongoDB <code>AuditLog</code> collection (No cryptographic chain)

3. What Have We NOT Built Yet? (Missing Features)

Based on the promises made in your PPT, the following features are missing from our current codebase:

1. **Offline-Ready Inspector App:** The PPT heavily promotes an offline-first mobile app that syncs data when online. Our current React frontend requires a continuous internet connection.
2. **GPS & Auto-Planned Routes:** The PPT mentions "Auto-planned routes" and GPS capabilities for LMOs. There is currently no geospatial logic or routing algorithm in the backend.
3. **Online Payments:** Slide 2 mentions traders can "Apply and pay online". We currently do not have any Payment Gateway integration (e.g., Razorpay/Stripe).
4. **SMS/WhatsApp Alerts:** Expiration alerts are currently handled via in-app dashboard badges/notifications, not external SMS APIs.

4. Irrelevant Content in the PPT (Immediate Fix Required)

[!CAUTION] Major Copy-Paste Error on Slide 2! At the very bottom of Slide 2, there is a quote that says: "India's first end-to-end traceable, community-powered waste solution—real rewards, real impact, real change."

Action: Delete this immediately. This is about waste management and is completely irrelevant to Legal Metrology and weighing scales.

5. What You Need to Add/Modify in the PPT

To make sure your presentation perfectly aligns with the impressive system we've built, please consider making the following adjustments to the PPT:

A. Fix the Tech Stack Slide (Slide 1)

If judges ask to see your code, the current Tech Stack slide will cause disqualification or point deduction for false claims. Update the icons to:

- **Frontend:** React, Vite, Tailwind CSS
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Security:** JWT (JSON Web Tokens)

Note: If you plan to say "This is our proposed architecture for scale, but the prototype is built on MERN", you MUST explicitly mention the word "Proposed Architecture" on that slide.

B. Highlight the "7-Point Statutory Inspection Engine"

Our codebase has a very robust **7-Point Digital Inspection Engine** (Visual condition, display readability, zero error, corner accuracy, calibration, sealing, and environment). **Action:** Add this to Slide 1 or Slide 2. It shows deep domain knowledge of Legal Metrology and is a massive selling point that you actually have working in code.

C. Clarify "Mobile App" vs "Mobile-Responsive Web App"

Since we built a web application, replace references of "Inspector Mobile App" with "Mobile-Responsive Inspector Portal". Alternatively, you can list the Mobile App under "Future Scope / Phase 2".

D. Rephrase Cryptographic Claims

Instead of "Hash-chained audit log" and "KMS/HSM", which we haven't built, use terms like:

- *"Immutable Digital Audit Trails"*
- *"Tamper-Evident QR Verification"*
- *"Role-Based Access Controls (RBAC)"*